Sougandhi Manonmani Pendyala

SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

Software engineer with Amazon experience building and operating **distributed systems** and full-stack features with Java, Spring Boot, React, TypeScript, and AWS. Re-architected a request-blocking flow into an event-driven SQS FIFO pipeline, carried on-call, and cut service cold-start time by 33%. Builds AI-backed products on the side, including SafeSend, PreVisit and Hearth.

CORE COMPETENCIES

Distributed Systems | Event-Driven Architecture | Java & Spring Boot | React & TypeScript | REST APIs & Microservices | AWS (SQS, DynamoDB, CDK) | Testing & CI/CD | On-Call & Reliability

WORK EXPERIENCE

Amazon

Sep 2025 - Oct 2026

Software Development Engineer

Detroit, MI, USA

- Eliminated blocked requests in offer expansion by re-architecting it from synchronous calls to an event-driven SQS FIFO pipeline handling **1,000+ messages/day**: built the ordered consumer and filter-chain routing, with DLQ recovery and tiered CloudWatch alarms provisioned via AWS CDK.
- Gave on-call a config-push lever to change rate limits across regions in minutes, work that previously required a full service deployment, by building policy-driven rate limiting and traffic shaping on AWS AppConfig, defined in code and CDK.
- Replaced an ops-mediated process with a self-service migration workflow for BIL (Build International Listings), using Spring Boot REST APIs and React with merchant access validation, migration-intent APIs, and a localized in-app banner; shipped with **100%** unit coverage in Vitest.
- Launched the Expansion Preferences page end to end in React with Zustand behind an A/B experiment, enabling self-service configuration of multiple offer expansion preferences and reducing turnaround time by **25%**; added a cooldown with localStorage persistence and polling hooks that reduced duplicate submissions by 40%.
- Cut Java microservice cold-start time by **33%** (90s to 60s) through JVM tuning, startup automation, and deployment-pipeline changes; carried on-call, resolving incidents, driving RCAs, and adding diagnostic logging and alarms that reduced time-to-diagnose by 30% for recurring failures.
- Halved feature turnaround from 2 weeks to 1 by building Claude and Kiro into 4 development workflows: implementation, test generation, debugging, and iteration, while maintaining production-quality code.
- Safely retired **2M** unproductive merchant connections with a dry-run removal script, after a 30-day communication campaign reaching 5M sellers.

PROJECTS

SafeSend

Built a hybrid AI moderation system that blocks abusive messages before they are sent: an on-device text normalizer and multilingual word list (~40 languages) cascades into a zero-shot, multi-label Claude classifier (20 categories) with JSON-schema structured output, prompt-injection defence, explainable reasons, and fail-closed handling; added human-in-the-loop appeals and a 41-message dialect-fairness evaluation.

JavaScript, Node.js, HTML, Claude API | LLM Evals, Guardrails, Prompt-Injection Defence

PreVisit

Built a patient-controlled health-records app for the Nebius x NVIDIA Global AI Hackathon: Nemotron vision and text models turn photographed medical reports into structured JSON with a source quote for every fact, patients verify each fact against the original, and doctors get an expiring, revocable link with an access log; turning off model reasoning took JSON parsing from 3 of 4 cut-off reports to every report parsing, with answers 3-5x shorter; deployed live on Render.

Python, FastAPI, SQLite, NVIDIA Nemotron, Nebius Token Factory | Structured Output, Human-in-the-Loop, Vision-Language Models

Hearth Move-In Guardian

Built 4 agents for the Meta Global AI Developer Hackathon that help renters keep their deposit: a lease checker that flags risky clauses with word-for-word quotes verified against the lease, move-in and move-out agents that outline damage with SAM 3.1 and compare photos side by side, and a deduction-dispute agent that challenges a charge only with verified evidence and drafts a point-by-point reply; code enforces every check, agents retry or mark findings unclear instead of guessing, and leases in Telugu and Hindi are supported, with an offline mock mode and a pytest suite.

Python, FastAPI, Meta Muse Spark, SAM 3.1 | AI Agents, Computer Vision, Multilingual

Mail Scheduler Agent

Built a Claude agent that schedules emails from plain English; Supabase Edge Functions + pg_cron send via Gmail with retries.

Python, FastAPI, Supabase, Claude, Gmail API | AI Agents, Tool Calling

Platepix3D

Built a prototype turning dish photos into interactive 3D menu models: Depth Anything V2 runs in the browser through Transformers.js (WebGPU or WASM) to build a photo-textured 3D mesh with .glb export, MobileNet classifies the dish, and react-three-fiber renders it; added inventory forecasting and a personalized menu ranker.

Next.js, TypeScript, Hugging Face Transformers.js, TensorFlow.js, Three.js, Tailwind | Computer Vision, Depth Estimation

Pawcare

Developed a pet care app for grooming, vet appointments, and pet-sitting with scheduling and reminders; integrated Google Maps, Firebase, and Calendar APIs.

Java, Android Studio

EDUCATION

Master of Information Technology and Management Illinois Institute of Technology — Aug 2023 - May 2025

Chicago, USA

CGPA: 4.0/4.0

Bachelor of Technology in Electronics and Communication Engineering Kamala Institute of Technology and Science — 2018 - 2022

Telangana, India

CGPA: 3.0/4.0

SKILLS

Languages: Java, TypeScript, JavaScript, Python, SQL, HTML, CSS

Full Stack: React, Redux, Zustand, Next.js, Node.js, Spring Boot, FastAPI, Tailwind CSS, Three.js, REST APIs, Microservices, Distributed Systems

Cloud & AWS: AWS CDK, SQS, DynamoDB, CloudWatch, IAM, AppConfig, Render, Nebius Token Factory

Data & Integrations: SQLite, Supabase, DynamoDB, Gmail API **Testing:** JUnit, Mockito, Vitest, pytest, Integration Testing, Functional Testing

DevOps & Tools: Docker, Kubernetes, CI/CD, Git, GitHub, GitLab, Jira, Claude, Kiro

AI & LLMs: Claude API, OpenAI API, NVIDIA Nemotron, Meta Muse Spark, SAM 3.1, Hugging Face Transformers.js, TensorFlow.js, Tool Calling, AI Agents, Prompt Engineering

AI Engineering: Structured Output (JSON Schema), LLM Evaluation (Evals), Guardrails & Prompt-Injection Defence, Human-in-the-Loop Review, Multimodal & Vision-Language Models, Computer Vision

Core CS: Data Structures, Algorithms, Object-Oriented Programming